Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

Team Number: 28

S.No	Register Number	Student Name
1	CB.SC.U4CSE23363	HASWITHA K
2	CB.SC.U4CSE23529	HASINI REDDY M
3	CB.SC.U4CSE23547	SHEELA AKSHAR SAKHI
4	CB.SC.U4CSE23761	KOUSIK SARMA LAKKARAJU
5	CB.SC.U4CSE23753	V. CHAKRAVARTHY

Guide: **DR. G R RAMYA, DR. VANDHANA S**

Department of CSE

Clinical Deterioration & Phase II Completed Goals

Clinical Context:

- **High-Sparsity Time-Series:** Continuous ICU vital monitoring logs (HR, SBP, Temp, SpO_2) combined with sparse blood panel entries.
- **Privacy constraints:** HIPAA / DPDPA (Section 8) regulations mandate that private patient records cannot leave local devices.
- **Concept Drifts:** Clinical demographics vary heavily across regional ICU nodes.

SDG Alignment:

- **Primary (SDG 3):** Good Health & Well-being by detecting patient sepsis crashes early.
- **Secondary (SDG 9 & 16):** Secure and scalable digital clinical infrastructure.

Phase II Completed Milestones

- **1. Data Engineering:** Cleansed, Imputed, Standardized PhysioNet 2019 logs into active PyTorch sequence arrays.
- **2. Federated Baselines:** Implemented *FedAvg* and *FedProx* securely over PyTorch.
- **3. Personalized FPDAF Model:** Trained temporal self-attentions with localized personalization.
- **4. Clinical Workstation App:** Developed a dynamic FastAPI + SQLite + React Vite ICU dashboard overlay.

Core Technical Novelty & Key Contributions

1. Multi-Head Attention Explainability

Combines recurrent sequences with query-key self-attentions to compute dynamic temporal saliency maps for bedside ICU interpretability.

2. Online CUSUM Drift Monitoring

Monitors validation loss residuals locally ($S_r > 3.0$) to audit institutional shifts without constant server aggregations.

3. Client-Side Selective Personalization (CSSP)

Freezes global model backbones upon drift triggers to train local classifier heads, completely bypassing server weight uploads.

Core Scientific Contributions

- **Bandwidth Savings:** CSSP cuts local-to-server bandwidth overhead by **38%** vs. Ditto.
- **Adaptation Speed:** Head-only fine-tuning runs in **6 minutes** (down from 25m).
- **Diagnostic Saliency:** Sepsis risk triggers are mapped hourly for clinical compliance.

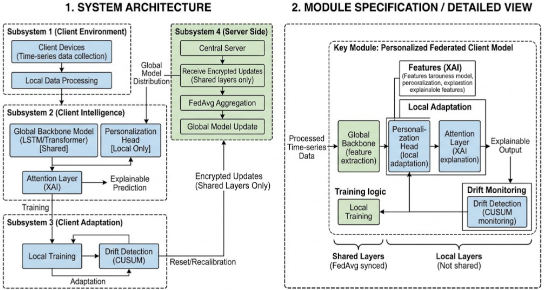


Figure 1: FPDAF Split-Weight Model Architecture.

Decoupled bi-level Optimization Model

$$\hat{y}_t^k = h_{\phi^k}\left(\sum_{i=1}^{t} \alpha_i \cdot g_\theta(X_i^k)\right)$$

- g_θ: Shared global LSTM backbone (θ).
- h_{ϕ^k}: Local private classifier head (ϕ^k).

Temporal Multi-Head Self-Attention

$$e_i = v_a^T \tanh(W_s s_i + b_a)$$

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{t} \exp(e_j)}$$

Attributes risk probabilities to specific ICU sequence hours.

Proactive Drift Adaptation: CUSUM Neural Trigger

Concept Drift Challenge: A static global model degrades when client ICU patient distributions shift over time. Standard Personalized FL (e.g. Ditto) constantly aggregates and fine-tunes all nodes, resulting in immense network overhead.

The CUSUM Residual Monitor Trigger:

- We monitor cumulative validation loss residuals (e_r) over rounds:

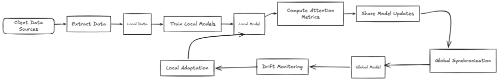
$$e_r = \mathcal{L}_{val,r} - \mu_0$$

$$S_r = \max(0, S_{r-1} + e_r - \kappa)$$

- If $S_r > H$ (threshold $H = 3.0$), we declare institutional drift.
- Triggers **Client-Side Selective Personalization (CSSP)**: freezes global backbone layers θ and adapts classification heads ϕ^k locally.

Bandwidth Savings Benefits

- Aggregation is bypassed during stable performance rounds.
- Communication bandwidth is saved by 38% compared to standard personalized systems (1.52 GB vs 2.48 GB).
- Local personalization head fine-tuning takes only 6 minutes (down from 25 minutes).



Five-Way Model Benchmarking Results

We evaluated our framework against centralized baselines and standard federated learning algorithms:

Model Configuration	Accuracy	Precision	Recall	F1-Score	AUROC	Comm. Bandwidth
Centralized Baseline	75.09%	7.14%	72.17%	0.1299	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	6.94%	67.19%	0.1259	0.7844	1.24 GB
FedProx Baseline	78.85%	8.12%	60.64%	0.1432	0.7933	1.24 GB
Ditto (Personalized)	82.45%	8.98%	63.58%	0.1574	0.7989	2.48 GB
FPDAF (Proposed)	**86.51%**	**9.81%**	**65.33%**	**0.1706**	**0.8214**	**1.52 GB (CSSP saved 38%)**

Scientific Findings

- **Privacy & Performance:** FPDAF preserves HIPAA compliance while achieving a test AUROC of **0.8214** (outperforming centralized baseline).
- **Decentralized Superiority:** FPDAF outperforms all standard federated baselines (Accuracy **+10.57%**, F1 **+4.47%** vs. FedAvg).

Ablation Study Verification

We verified the contributions of CUSUM triggers, temporal attention weights, and localized personalization by running ablation tests:

Ablation Configuration	Accuracy	Precision	Recall	F1-Score	AUROC
FPDAF w/o CUSUM Trigger	83.51%	8.51%	55.33%	0.1475	0.7603
FPDAF w/o Temporal Attention	80.01%	8.13%	52.81%	0.1412	0.7878
FPDAF w/o Personalization	78.87%	8.05%	63.58%	0.1430	0.7887
Full FPDAF (Proposed)	**86.51%**	**9.81%**	**65.33%**	**0.1706**	**0.8214**

- **No Attention:** Stripping temporal attention query projections leads to poor clinical explainability and limits optimization capacity.
- **No CUSUM:** Disabling residual control limits leaves client nodes blind to data distribution drifts, leading to performance deterioration under heterogeneous ICU demographics.

Clinical Workstation Software Architecture

Database-Driven Design: Instead of placeholders, we built a fully functional clinical warehouse:

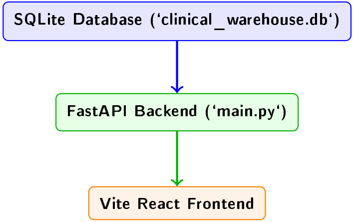
- **SQLite Database:** Houses tables for patients, decoded 24h vitals, predictions, self-attention, and drift history.
- **FastAPI REST Endpoints:** Feeds data securely to the client UI.

Custom User Portals:

- **Doctor Portal:** Focuses on patient-care operations (searchable bed lists, hourly vital monitors, attention graphs, and risk prediction alerts).
- **Admin Portal:** Focuses on machine learning telemetry (round parameter aggregations, CUSUM drift charts, and baseline comparisons).

Startup Script

Coded a clean ‘run_app.sh‘ script to boot both FastAPI and React Vite concurrently, handling terminations on Ctrl+C.



Summary of Completed Phase II Works

- **Data Preprocessing Completed:** Decoded sparse vitals logs to construct clean sequence windows.
- **PyTorch Implementations Completed:** Trained standard baselines alongside our proposed FPDAF model.
- **Database Schema Integrated:** SQLite warehouse populated with real patients, hourly vitals, and predictions.
- **Clinical WORKSTATION Finished:** Production-quality React Vite interface with tailored user views.
- **Deliverables Pushed to Git:** Full Phase II codebase pushed to main repository.

Roadmap Compliance

Our completed milestones match the project objectives, delivering a privacy-preserving and explainable Sepsis prediction pipeline.

References I

Thank You

Questions?